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Interpretable Modelling of Greenhouse Environment through SINDy Approach

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Abstract
Modelling of greenhouse environment is crucial for maintaining the optimal environment for crop growth. Digital twins are widely used in greenhouse modeling due to their real-time reflection of reality. However, most existing Digital twins for greenhouse are based on data-driven models. This method is highly dependent on the dataset and lacks interpretability and generalizability. In this study, a knowledge-infused data-driven model was built by using sparse identification of nonlinear dynamical systems (SINDy). We collected 1927 sets of environmental data from a solar greenhouse in the cold region of China during the period from 22 February to 28 February 2024, and identified the predictive model from the measured data and physical knowledge. Comparing with the data-driven model, the SINDy model is more interpretable. Comparing with the physics-based model, the SINDy model is more concise and accurate, with the mean absolute percentage error (MAPE) of only 5.7% from the measured values. This has important implications for greenhouse control.

Keywords
Interpretable modelling; Sparse Identification of Nonlinear Dynamics System; Greenhouse Environment;

1 Introduction

Greenhouse environmental control plays a pivotal role in maintaining optimal crop growth (Svensen *et al*. 2023). Research has shown that crop growth is tightly related to temperature and humidity. Both excessively high and low temperatures can cause crop growth to stop and yields to decreases significantly (Ahamed *et al*. 2019). High humidity could lead to fungal diseases, leaf necrosis, and calcium deficiencies. And also low humidity can cause crop to close its stomatal openings and slow down the photosynthesis rate, leading to decreased yields (Chen *et al*. 2022). From this, it can be seen that appropriate temperature and humidity can increase crop production. However, temperature and humidity in greenhouse are affected by various factors, such as sudden changes of weather conditions, failure and ageing of equipment and inexperienced managers, which cause the uncertainty, complexity, and strong coupling of the greenhouse system. This poses significant challenges for the control of the greenhouse environment.

Model predictive control (MPC) is a suitable approach to control such a multi-input multi-output system with hard control constraints (Mayne *et al*. 2014). It has been proved that MPC is superior to other classical control methods (Shang *et al*. 2019). With the popularization of the Internet of Things (IoT), the cost of data acquisition has decreased, leading to increased attention on data-driven model predictive control (DMPC). Mahmood et al. proposed a DMPC approach based on artificial neural networks (ANNs), which offers higher prediction accuracy compared to traditional predictive models, thus providing a solid foundation for subsequent stable control (Mahmood *et al*. 2023). Chen et al. introduced a DMPC method that combines principal component analysis (PCA) and kernel density estimation (KDE). This approach exhibits lower control costs compared to rule-based control (RBC) and other model predictive control methods (Chen *et al*. 2021). Hu et al. found that in greenhouse control, DMPC not only consumes less energy but also enhances crop yields (Hu *et al*. 2022). As can be seen, DMPC exhibits higher accuracy, lower control costs, and more efficient practical applications. However, since DMPC is based on data-driven modeling, the model is highly dependent on data, leading to poor generalizability. Additionally, the model is opaque, lacking interpretability in its outputs, which makes it difficult for users to trust the decisions made by the system.

This paper proposes a digital twin framework that employs the sparse identification of nonlinear dynamics (SINDy) technique to establish predictive models for greenhouse control. SINDy is a cutting-edge modeling technique rooted in sparse regression and compressive sensing principles. Compared to traditional data-driven models, SINDy incorporates prior knowledge and physical laws during the modeling process, effectively combining data-driven approaches with physical principles. This integration significantly enhances the interpretability and stability of the models, which will substantially improve the reliability of predictive control strategies in greenhouse environments. The remainder of this paper is organized as follows: Section 2 introduces the digital twin framework and the SINDy modeling method. Section 3 describes our experiments; Section 4 presents the comparison of experimental results and discussions; Section 5 concludes the paper.

2 Method

2.1 Digital Twin Framwork

The digital twin theoretical framework is divided into four layers (Verdouw *et al*. 2021) in this paper: (1) Device layer: Hardware components used to attach physical objects and interact directly with them, such as sensors and control actuators. (2) Network Layer: Components used for efficient and secure communication, such as internet gateways and cloud proxy machines. (3) Integration Layer: Components used for data processing and integration, including data storage and data management. (4) Application Layer: Functions that provide specific capabilities to users based on the information accessed from the integration layer, such as machine learning and MPC.

In the digital twin framework, the device layer collects real-time building operational data through deployed sensors. The network layer ensures reliable data transmission based on industrial IoT protocols such as MQTT and OPC UA. The data is then transferred through the network layer to the integration layer for preprocessing, standardized storage, and management. The application layer accesses data from the integration layer through reserved API interfaces, enabling building energy consumption prediction, operational state simulation, and 3D visualization display by incorporating machine learning algorithms.

Since the implementation of the device layer, network layer, and integration layer largely depends on technology and equipment, this paper focuses solely on the application layer. The core of the application layer is the construction of the prediction model. Kaiser et al. conducted a systematic experimental study comparing the performance of the SINDy model and neural network models in terms of prediction accuracy and control performance. The results demonstrate that the SINDy model not only significantly outperforms neural network models in performance but also exhibits substantial advantages in efficiency, with its execution speed being 37 times faster than that of neural network models (Brunton *et al.* 2016).

2.2 Construction of Predictive Model

Sparse identification of nonlinear dynamical systems, proposed by Brunton et al., is based on data-driven methods. Physical knowledge are incorporated into the modeling process by constructing candidate function library. In recent years, sparse identification has been widely used for model identification of complex dynamic nonlinear systems in various fields (Abdullah *et al.* 2023; Liu *et al.* 2023; Im *et al.* 2022).

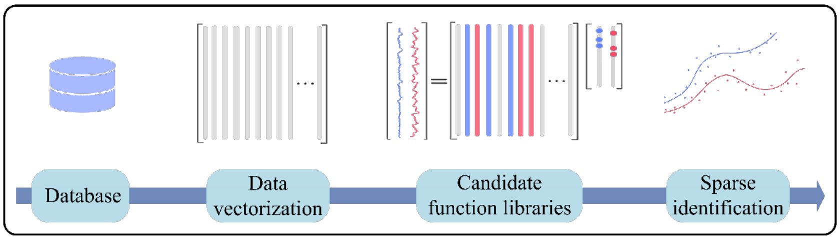

Figure 1. Implementation steps for SINDy

The implementation steps of model construction is shown in Figure 1. SINDy is used to construct the thermal and humidity environment prediction model of greenhouse. The description of the dynamic system can be given by the following equation:

$$\dot{X} = F(X) \qquad (1)$$

where $X \in R_n$ represents the state, $\dot{X}$ represents the trend of the system, and F indicates the continuous nonlinear state dynamics.

In the data processing procedure, a matrix structure is employed for data organization, where the feature dimensions are represented as column vectors of the matrix X, while the time series dimension is represented as row vectors. The left-hand side of *Eq.(1)* can be approximated by measurement or numerical derivatives of the state variables. The right-hand side of *Eq.(1)* can be estimated based on a library of candidate function $\Theta(X)$.

$$\Theta(X) = \begin{bmatrix} | & | & \cdots & | & | \\ 1 & X & \cdots & \sin X & \cos X \\ | & | & \cdots & | & | \end{bmatrix} \qquad (2)$$

Based on existing research in the field and prior physical system knowledge, this study adopts a systematic function selection approach to construct a candidate function library comprising monomials, polynomials, and trigonometric functions. Once the library is constructed, set the weight coefficients ξ for each term, and all coefficients together form the matrix $\Xi \in R_n$. The state equation can be expressed as:

$$\dot{x} = \Xi \cdot \Theta(X) \quad (3)$$

At this point, with the prediction model *Eq.(3)* and the measured values, the identification of the prediction model is transformed into a regression problem. This process is usually solved by sparse regression, i.e., least squares and L_2 regularization:

$$\hat{\Xi} = \arg\min_{\Xi} \left\| \dot{X} - \Theta(X) \cdot \Xi \right\|_2^2 + \lambda \left\| \Xi \right\|_2^2 \quad (4)$$

To ensure the simplicity of the state equation, a parameter threshold ε is set, coefficients less than ε are set to zero, and then the coefficient matrix Ξ is repeatedly substituted into *Eq.(4)* until the matrix Ξ converges. Ultimately, the prediction model is obtained:

$$\dot{X} = \hat{\Xi} \cdot \Theta(X) \quad (5)$$

3 Case Study

In this paper, the experiment was conducted in a greenhouse in cold region of China. The greenhouse is 5*m* wide and 8*m* high. The main structure is made of hot-dip galvanized rectangular tubes, and the walls are made of 5*mm* tempered glass. The roof is covered with 5*mm* polycarbonate sheet. The temperature and humidity of greenhouse is determined by multiple factors. In this paper, outdoor temperature (T), indoor temperature (t), outdoor humidity (H), indoor humidity (h), outdoor wind speed (W) and solar radiation (S) were selected to represent the state of the greenhouse system.

3.1 Data Collection

In the experiment, measurement data were obtained by placing measurement instruments inside and outside the greenhouse. The height of the equipment was uniformly set to 1.2*m*, and the location is shown in Figure 2. The sensor models and parameters used are shown in Table 1.

3.2 Data Processing

The data were sampled from 22th February 2024 to 28th February 2024, with sampling intervals set at 5*min*, and 1927 sets of data were collected. Eventually, the instrument data were imported into the computer.

Due to the inertia of temperature and humidity, they do not change abruptly in short time. Therefore, moving average filtering was used to process the anomalous data to fit the changing characteristics of temperature and humidity.

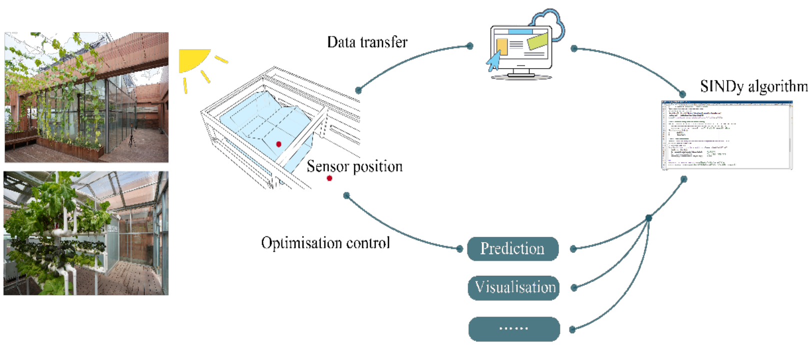

Figure 1. The procedure of experiment

Table 1. Parameters related to environmental measurement instruments

Instrument name	Measurement type	Accuracy	Interval	Picture
Kestrel Meter 5400 Heat Stress Tracker	Wind speed, temperature, humidity	±3%, ±0.5℃, 2%RH	5*min*	
HOBO Solar Radiation Smart Sensor	solar radiation	±10W/m^2	5*min*	

3.3 Model Identification

The indoor humidity/indoor temperature at moment $k+1$ was taken as the output, outdoor temperature (T), indoor temperature (t), outdoor humidity (H), indoor humidity (h), outdoor wind speed (W) and solar radiation (S) at moment k was taken as the input. The functional relationship between them was recognized directly by SINDy. Based on the physical laws and priori knowledge of heat and moisture transfer in the building envelope, a library of candidate function Θ(X) was built:

$$\Theta(X_k)=\begin{bmatrix} | & | & | & | & | & | & | & | & | \\ 1 & X_k & \sin X_k & \cos X_k & X_k\cdot X_k & \sin^2 X_k & \cos^2 X_k & X_k\cdot\sin X_k & X_k\cdot\cos X_k \\ | & | & | & | & | & | & | & | & | \end{bmatrix} \quad (6)$$

Built the sparse coefficient matrix Ξ. the loss function was solved by the least squares:

$$\mathrm{J}(\Xi)=\arg\min_{\Xi}\sum_{k=1}^{m}\sqrt{\left\|X_{k+1}-\Theta(X_k)\cdot\Xi\right\|_2^2} \quad (7)$$

Set up the array λ=[ε_1, ε_2, … ε_n], selected the threshold ε_n in turn, the sparse coefficients less than ε were set to zero, and then the coefficient matrix Ξ_n was repeatedly substituted into *Eq.(7)* until the preconditions are satisfied.

4 Result and Discussion

4.1 Modelling Results

The dataset was divided into training and validation sets at 4:1. The training set was utilized for model parameter optimization and learning, while the validation set served to evaluate the model's generalization performance. The parameters of the algorithm were adjusted according to the verification results and the final prediction models obtained are as follow:

$$\begin{aligned} t_{k+1} &= 1.1T_k + 5.2H_k - 2.6H_k\cos(h_k) + 3.1h_k\cos(H_k) - 0.08h_k\sin(T_k) \\ &- 1.4H_k\cos^2(h_k) - 4.3h_k\cos^2(H_k) - 3.9h_k\sin^2(H_k) - 1.3h_k\sin^2(h_k) \end{aligned} \quad (8)$$

$$\begin{aligned} h_{k+1} &= 0.4t_k - 1.3T_k + 1.7h_kT_k - 2.6h_kt_k + 0.3T_k\cos(t_k) + 1.8T_k\cos(H_k) \\ &- 1.1t_k\cos(H_k) + 1.1H_k\cos(H_k) - 0.9T_k\sin(h_k) + 2.1t_k\sin(h_k) \end{aligned} \quad (9)$$

Where T_k, t_k, H_k, h_k is the outdoor temperature, indoor temperature, outdoor humidity and indoor humidity at moment k; t_{k+1}, h_{k+1} is the indoor temperature and indoor humidity at moment $k+1$. As can be seen from *Eq.(8)* and *Eq.(9)*, the indoor temperature at moment $k+1$ in the experimental greenhouse is strongly correlated with the outdoor temperature, outdoor humidity, and indoor humidity at moment k. The indoor humidity at moment $k+1$ is also strongly correlated with the outdoor temperature, indoor temperature, outdoor humidity, and indoor humidity at moment k. The outdoor wind speed and solar radiation intensity have a relatively minor impact on the indoor temperature and humidity of the greenhouse.

The comparison of the predicted and measured values is shown in Figure 3. The predicted values maintain the general trend with the actual values. The change in temperature trend is based on days and peaks at midday accompanied by daylight. The fourth day was raining with low temperature. The humidity trend is also based on days, with daylight leading to transpiration followed by the increase of humidity. On the fourth day it peaked with the increase of atmospheric moisture.

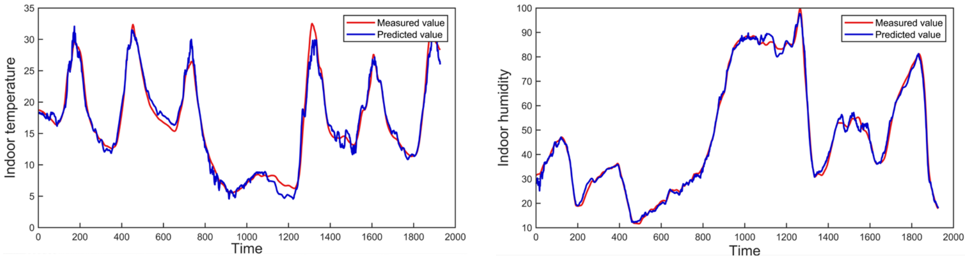

Figure 2. Comparison of the predicted and measured values

4.2 Model Comparison

The thermal equilibrium equation for the greenhouse system can be formulated as:

$$Q_r = Q_t + Q_s + Q_v \qquad (10)$$

Where Q_r represents the solar radiation energy transmitted into the greenhouse(W), Q_t denotes the heat transfer through the building envelope to the outside via radiation, conduction, and convection, referred to as the overall heat loss(W), Q_s indicates the portion of heat absorbed by the ground within the greenhouse, known as the ground heat transfer(W), Q_v represents the heat loss through building apertures (such as window and door gaps), termed as the air leakage heat loss(W). When $Q_r > Q_t + Q_s + Q_v$, the greenhouse accumulates heat, resulting in an increase in indoor temperature; conversely, the indoor temperature decreases. The corresponding calculation formulas are presented below:

$$Q_r = rSF_c \sin\theta \qquad (11)$$

$$Q_t = \sum k_a F_a (t - T) \qquad (12)$$

$$Q_s = \sum k_d F_d (t - T) \qquad (13)$$

$$Q_v = nVC_n p(t - T) / 3.6 \qquad (14)$$

Where r is transmission coefficients, S is solar irradiance level(W/m^2), θ is solar incidence angles, F_c is the light-receiving area(m^2), k_a is cross-flow heat release coefficients($W/m^2 \cdot ℃$), k_d is ground heat transfer coefficients($W/m^2 \cdot ℃$), F_a is the surface area of the greenhouse(m^2), F_d is the floor area(m^2), t is the indoor temperature, T is the outdoor temperature, n represents the air exchange rate per hour, calculated as 1.5 for airtight conditions, V denotes the greenhouse volume(m^3), C_n indicates the specific heat capacity of air($kJ/kg \cdot ℃$), p signifies the bulk density of outdoor air(kg/m^3).

Based on *Eq.(10)-Eq.(14)*, the white-box prediction model is shown below:

$$t_{k+1} = \frac{\left[rSF_c \sin\theta - (k_a F_a + k_d F_d + 67C_n p)(t_k - T_k) \right] \cdot H}{M \cdot C_n} + t_k \qquad (15)$$

Where H is the duration of exposure to light(s), M is the weight of the air in the greenhouse(kg), t_{k+1} is the indoor temperature of the next moment(℃), t_k is the indoor temperature at the present time(℃), and T_k is the outdoor temperature at the present time(℃).

The predicted values of the white-box and SINDy model are shown in Figure 4. The prediction accuracy of the SINDy model is significantly higher. This is because in the white-box model the reality of plant growth cannot be quantified, and some coefficients， such as solar incidence angles and light-receiving area, are too difficult to measure and can only be replaced by estimates. Moreover, the mean absolute error (MAE) of the white box model was 2.27°C and the mean absolute percentage error (MAPE) was 14.3%, while the MAE of the SINDy model was only 0.82°C and the MAPE was only 5.7%. Remarkably, the SINDy algorithm identifies the predictive model in only 7s, which means that the model can predict the state of the greenhouse in almost real-time. This is significant for the control of greenhouse.

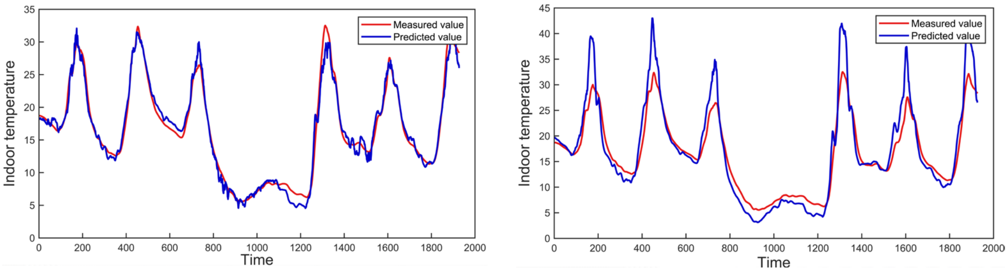

Figure 3. Comparison of the SINDy model (left) and white-box model (right)

In applications, the SINDy model filters out the key factors affecting the system and reduces the interference of redundant information. In contrast to the white-box model, SINDy model is more convenient to construct and more concise. In contrast to the data-driven model, the SINDy model can clearly demonstrate the factors and weights of the inputs that influence the output, which is more interpretable. However, SINDy models also have deficiencies: (1) Their sparsity may cause the model to lose some important but uncommon information, (2) The model requires some domain knowledge, (3) The appropriate model parameters are difficult to set, such as suitable hyperparameters and sparse coefficient thresholds.

5 Conclusions

The results of this study demonstrate the advantages of SINDy models in predicting the temperature and humidity within greenhouse. In the task of greenhouse environment prediction, the SINDy model demonstrates exceptional predictive performance: the Mean Absolute Error (MAE) is as low as 0.82℃, and the Mean Absolute Percentage Error (MAPE) is merely 5.7%. Furthermore, the model exhibits efficient model identification capabilities, significantly outperforming traditional prediction methods. While data-driven models have proven successful in past studies, the SINDy algorithm addresses their shortcomings by creating a more concise and interpretable model.

Our research has limitations that are important to point out. Although we have conducted an indepth exploration of the modeling aspects of the digital twin framework, the research primarily focuses on the model level and does not cover other components of the framework. However, as the core technical part of the entire framework, the feasibility of the model has been validated. This suggests that MPC strategies based on machine learning can possess high interpretability, thereby gaining user trust, and also exhibit good accuracy.

This paper successfully established a digital twin prediction model using SINDy technology. The model combines data-driven approaches with physical knowledge, not only enhancing the interpretability of the model but also surpassing traditional models in terms of simplicity, interpretability. These advantages can better enable intelligent control of the greenhouse environment, optimizing crop growth conditions and improving agricultural production efficiency. Moreover, the model's efficient construction and updating capabilities are of significant practical importance for achieving real-time data interaction and synchronization in greenhouse digital twin systems. Future research will focus on establishing a comprehensive experimental platform for greenhouse digital twins. Through field testing, we aim to validate the performance of the SINDy model in actual agricultural production environments, thereby providing technical support and practical references for promoting agricultural intelligent transformation and achieving sustainable development goals.

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